

8086 Microprocessor in Maximum Mode

Q. Explain the operation of the 8086 microprocessor in Maximum Mode. Discuss the role of the Bus Controller (8288) and how a multiprocessor environment is managed.

****Definition to Remember****

The 8086 microprocessor systems manage external devices.

1. Maximum Mode Configuration

In this mode, pins that previously generated control signals (like ``ALE``, ``WR``) take on new functions related to bus arbitration and multiprocessor coordination.

* The 8086 manages external devices.

2. Reassigned Maximum Mode Signals

* **`S2'`, `S1'`, `S0'` (Status Lines):** The 8086 outputs its current cycle status on these 3 pins. The 8288 reads them to generate command signals.

* **`RQ'`** (Receive/Transmit): The 8086 uses this signal to manage bus arbitration.

3. Role of the 8288 Bus Controller

The 8288 decodes the ``S2'-S0'`` signals to generate system controls:

* **`000`** (Active Low): Asserted when the 8086 is in `wait` state.

4. Managing the Multiprocessor Environment

* **Bus Arbitration:** To prevent data corruption, only one processor can use the bus at a time. A coprocessor asserts **`RQ'`**. The 8086 finishes its cycle, floats its buses (High-Z), and asserts **`GT'`** to hand over control.

* **`S2'`** (Status Line): The 8086 outputs its current cycle status on this pin. The 8288 reads it to manage bus arbitration.

****Must-Write Points (for 10 marks)****

1. **Maximum Mode Configuration**

*****Quick Recall*****

***`Maxim
RQ/GT***